Experiment No. 10

Title:

Malware Detection and Security Analysis Using ClamAV and Online Threat Intelligence Tools in Kali Linux

Aim:

To demonstrate malware detection using antivirus software and analyze suspicious files using threat intelligence tools.

Objectives:

After completing this experiment, students will be able to:

1. Install and configure antivirus software in Linux.
2. Update malware signature databases.
3. Scan files and directories for malware.
4. Detect malicious files using antivirus engines.
5. Verify malware using threat intelligence platforms.
6. Understand real-world malware analysis workflow.

Theory:

1 Antivirus and Malware Detection

Antivirus software detects malware using several techniques:

Technique	Description
Signature-based detection	Compares file patterns with known malware
Heuristic detection	Detects suspicious behavior
Behavioral analysis	Observes program actions
Threat intelligence	Uses global malware databases

2 ClamAV Antivirus

ClamAV

ClamAV is an open-source antivirus engine widely used for:

- Linux servers
- Email scanning
- Security gateways
- SOC threat detection systems

Features:

- Real-time malware detection
- Signature database updates
- Directory scanning
- Integration with automation tools

3 Threat Intelligence Platforms

Threat intelligence platforms help security analysts determine whether a file is malicious.

One popular platform is:

VirusTotal

VirusTotal analyzes suspicious files using **multiple antivirus engines simultaneously**.

Capabilities:

- Detect malware signatures
- Provide threat reputation
- Show community analysis
- Identify malicious URLs

4 Safe Malware Testing

To safely demonstrate malware detection, we use the **EICAR test malware file**.

EICAR Test File

Characteristics:

- Harmless file
- Recognized by all antivirus systems
- Used globally for testing antivirus functionality

System Requirements :

Component	Requirement
Operating System	Kali Linux
RAM	4 GB
Internet	Required
Antivirus	ClamAV

Installation of ClamAV :

Open Terminal.

Step 1 Update System

sudo apt update

Explanation
Updates package repository.

Step 2 Install ClamAV

sudo apt install clamav clamav-daemon -y

Explanation
Installs ClamAV scanning engine and daemon service.

Step 3 Verify Installation

clamscan --version

Example output

ClamAV 1.x.x

Configuration :

Step 1 Update Virus Signatures

(Before Step 1 use following command)

sudo systemctl stop clamav-freshclam

sudo systemctl stop clamav-daemon

sudo freshclam

Explanation
Downloads latest malware signatures.

Example output

Database updated

Step 2 Start ClamAV Service

sudo systemctl start clamav-daemon

Verify service

sudo systemctl status clamav-daemon

Malware Detection Demonstration

Step 1 Create Lab Directory

mkdir malware_lab

cd malware_lab

Step 2 Create Normal File

echo "This is a normal file" > normal.txt

Step 3 Create Test Malware

echo "X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*" > eicar.txt

echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.com

Explanation
This creates the **EICAR malware test file**.

Step 4 Scan Files

clamscan malware_lab

Expected output

eicar.txt: Eicar-Test-Signature FOUND

Step 5 Recursive Scan

clamscan -r malware_lab

Explanation
Scans all files inside directory recursively.

Step 6 Remove Malware

clamscan --remove malware_lab

Output

Removed eicar.txt

Threat Intelligence Verification:

Now verify malware using **VirusTotal**.

Steps:

1. Open browser.
2. Visit VirusTotal website.
3. Upload **eicar.txt** file.

Result:

- Multiple antivirus engines detect it as malware.

Explanation for students:

This shows how **security analysts verify suspicious files using global threat intelligence databases**.

Real World SOC Workflow Demonstration

Instructor explains workflow:

User downloads suspicious file

↓

Antivirus scans file

↓

Malware detected

↓

File analysed using threat intelligence

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Security team blocks threat

This demonstrates **real cybersecurity operations workflow**.

Expected Output

Example scan summary:

----------- SCAN SUMMARY -----------

Known viruses: 8700000+

Scanned files: 2

Infected files: 1

Observations:

Activity	Result
ClamAV installed	Successful
Signature update	Completed
Normal file scan	Clean
EICAR file scan	Malware detected
Threat intelligence verification	Malware confirmed

Result:

ClamAV successfully detected malware and the file was verified using threat intelligence tools.

Conclusion:

The experiment demonstrated the working of antivirus software using ClamAV. Malware was detected using signature-based detection and verified using online threat intelligence tools.
